6. Write the python program for Vacuum Cleaner problem

Aim

To develop a Python program to simulate the Vacuum Cleaner Problem in Artificial Intelligence by cleaning all dirty rooms using a simple reflex agent.

Algorithm

1. Start the program.
2. Read the status of Room A and Room B from the user (Dirty/Clean).
3. Read the initial location of the vacuum cleaner.
4. If the current room is dirty, clean it.
5. Move to the other room.
6. If the other room is dirty, clean it.
7. Display the action performed in each room.
8. Display the final status of both rooms.
9. Stop.

Program

```
# Vacuum Cleaner Problem

# Input room status
roomA = input("Enter status of Room A (Dirty/Clean): ").capitalize()
roomB = input("Enter status of Room B (Dirty/Clean): ").capitalize()

# Input vacuum location
location = input("Enter Vacuum Location (A/B): ").upper()

print("\nInitial State:")
print("Room A:", roomA)
print("Room B:", roomB)
print("Vacuum Location:", location)

# Cleaning process
if location == "A":
```

```
    if roomA == "Dirty":
        print("\nCleaning Room A...")
        roomA = "Clean"

    print("Moving to Room B...")

    if roomB == "Dirty":
        print("Cleaning Room B...")
        roomB = "Clean"

elif location == "B":
    if roomB == "Dirty":
        print("\nCleaning Room B...")
        roomB = "Clean"

    print("Moving to Room A...")

    if roomA == "Dirty":
        print("Cleaning Room A...")
        roomA = "Clean"

else:
    print("Invalid Vacuum Location!")

print("\nFinal State:")
print("Room A:", roomA)
print("Room B:", roomB)
print("\nAll rooms are clean.")
```

Input

```
Enter status of Room A (Dirty/Clean): Dirty
Enter status of Room B (Dirty/Clean): Clean
Enter Vacuum Location (A/B): A
```

Output

```
Initial State:
Room A: Dirty
Room B: Clean
Vacuum Location: A

Cleaning Room A...
Moving to Room B...

Final State:
Room A: Clean
Room B: Clean

All rooms are clean.
>>> |
```

Result

The Python program was successfully implemented to simulate the Vacuum Cleaner Problem. The program accepts the initial status of Room A and Room B along with the vacuum cleaner's initial location, cleans all dirty rooms, and displays the sequence of actions and the final state where both rooms are clean.